

CLIMATE CLASSIFICATION (KOPPEN) / INDIA CLIMATIC REGIONS

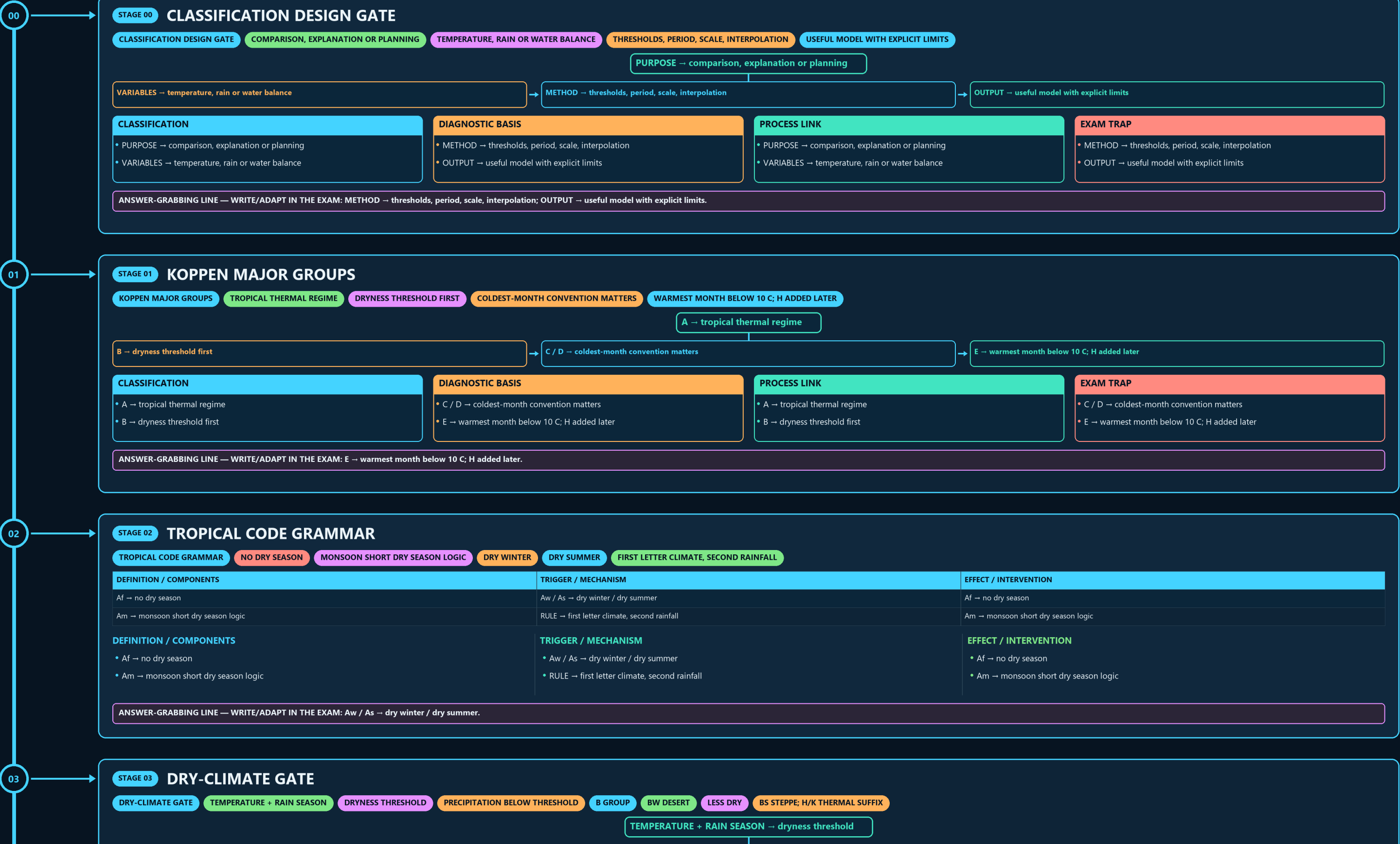
CLASSIFICATION DESIGN GATE → KOPPEN MAJOR GROUPS → TROPICAL CODE GRAMMAR → DRY-CLIMATE GATE → C-D THRESHOLD FIREWALL → NORMAL-TO-TREND LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

DEFINITION / COMPONENTS

- Af → no dry season
- Am → monsoon short dry season logic

TRIGGER / MECHANISM

- Aw / As → dry winter / dry summer
- RULE → first letter climate, second rainfall

EFFECT / INTERVENTION

- Af → no dry season
- Am → monsoon short dry season logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aw / As → dry winter / dry summer.

03

STAGE 03

DRY-CLIMATE GATE

DRY-CLIMATE GATE

TEMPERATURE + RAIN SEASON

DRYNESS THRESHOLD

PRECIPITATION BELOW THRESHOLD

B GROUP

BW DESERT

LESS DRY

BS STEPPE; H/K THERMAL SUFFIX

TEMPERATURE + RAIN SEASON → dryness threshold

PRECIPITATION BELOW THRESHOLD → B group

DRIER → BW desert

LESS DRY → BS steppe; h/k thermal suffix

DEFINITION / COMPONENTS

- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

TRIGGER / MECHANISM

- DRIER → BW desert
- LESS DRY → BS steppe; h/k thermal suffix

EFFECT / INTERVENTION

- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LESS DRY → BS steppe; h/k thermal suffix.

04

STAGE 04

C-D THRESHOLD FIREWALL

C-D THRESHOLD FIREWALL

COLDEST MONTH

C/D DISCRIMINATOR

CONVENTION 1

MINUS 3 C BOUNDARY

CONVENTION 2

0 C BOUNDARY

ANSWER RULE

DEFINITION / COMPONENTS

COLDEST MONTH → C/D discriminator

CONVENTION 1 → minus 3 C boundary

TRIGGER / MECHANISM

CONVENTION 2 → 0 C boundary

ANSWER RULE → state convention before mapping

EFFECT / INTERVENTION

COLDEST MONTH → C/D discriminator

CONVENTION 1 → minus 3 C boundary

DEFINITION / COMPONENTS

- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

TRIGGER / MECHANISM

- CONVENTION 2 → 0 C boundary
- ANSWER RULE → state convention before mapping

EFFECT / INTERVENTION

- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVENTION 2 → 0 C boundary; ANSWER RULE → state convention before mapping.

05

STAGE 05

NORMAL-TO-TREND LADDER

NORMAL-TO-TREND LADDER

STATED 30-YEAR REFERENCE

OBSERVATION MINUS NORMAL

SPREAD AND RECURRENCE

MULTI-PERIOD DIRECTION WITH UNCERTAINTY

NORMAL → stated 30-year reference

ANOMALY → observation minus normal

VARIABILITY → spread and recurrence

TREND → multi-period direction with uncertainty

DEFINITION / COMPONENTS

- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

TRIGGER / MECHANISM

- VARIABILITY → spread and recurrence
- TREND → multi-period direction with uncertainty

EFFECT / INTERVENTION

- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06

CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODEL

MEASURED VARIABLES

QUALITY CONTROL

COMPARABLE RECORDS

INTERPOLATION + THRESHOLDS

CLASS MAP

GROUND REALITY

TRANSITION, NOT WALL

STATIONS / GRIDS → measured variables

QUALITY CONTROL → comparable records

INTERPOLATION + THRESHOLDS → class map

GROUND REALITY → transition, not wall

DEFINITION / COMPONENTS

TRIGGER / MECHANISM

EFFECT / INTERVENTION

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Normal → based on year reference

• ANOMALY → observation minus normal

Variable → spread and recurrence

• TREND → multi-period direction with uncertainty

Normal → based on year reference

• ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODELMEASURED VARIABLESQUALITY CONTROLCOMPARABLE RECORDSINTERPOLATION + THRESHOLDSCLASS MAPGROUND REALITYTRANSITION, NOT WALL

STATIONS / GRIDS → measured variables

QUALITY CONTROL → comparable records

INTERPOLATION + THRESHOLDS → class map

GROUND REALITY → transition, not wall

DEFINITION / COMPONENTS

• STATIONS / GRIDS → measured variables

• QUALITY CONTROL → comparable records

TRIGGER / MECHANISM

• INTERPOLATION + THRESHOLDS → class map

• GROUND REALITY → transition, not wall

EFFECT / INTERVENTION

• STATIONS / GRIDS → measured variables

• QUALITY CONTROL → comparable records

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERPOLATION + THRESHOLDS → class map; GROUND REALITY → transition, not wall.

07

STAGE 07INDIA CONTROL MATRIX

INDIA CONTROL MATRIXLATITUDE + ALTITUDETHERMAL STRUCTUREUPLIFT, RAIN SHADOW, COLD BARRIERSEA DISTANCECONTINENTAL RANGEMONSOON + WDS + OCEANSEASONAL VARIABILITY

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
LATITUDE + ALTITUDE → thermal structure	SEA DISTANCE → maritime / continental range	LATITUDE + ALTITUDE → thermal structure	SEA DISTANCE → maritime / continental range
RELIEF → uplift, rain shadow, cold barrier	MONSOON + WDS + OCEAN → seasonal variability	RELIEF → uplift, rain shadow, cold barrier	MONSOON + WDS + OCEAN → seasonal variability

CLASSIFICATION

• LATITUDE + ALTITUDE → thermal structure

• RELIEF → uplift, rain shadow, cold barrier

DIAGNOSTIC BASIS

• SEA DISTANCE → maritime / continental range

• MONSOON + WDS + OCEAN → seasonal variability

PROCESS LINK

• LATITUDE + ALTITUDE → thermal structure

• RELIEF → uplift, rain shadow, cold barrier

EXAM TRAP

• SEA DISTANCE → maritime / continental range

• MONSOON + WDS + OCEAN → seasonal variability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MONSOON + WDS + OCEAN → seasonal variability.

08

STAGE 08INDIA CODE-MAP BOUNDARY

INDIA CODE-MAP BOUNDARYCOMMON TYPESTROPICAL, DRY, SUBTROPICAL, HIGHLANDEXACT CODEDEPENDS ON DATASET AND THRESHOLDDPENDS ON SCALE AND INTERPOLATIONCITE SOURCE, PERIOD AND CONVENTION

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
COMMON TYPES → tropical, dry, subtropical, highland	BOUNDARY → depends on scale and interpolation	COMMON TYPES → tropical, dry, subtropical, highland	BOUNDARY → depends on scale and interpolation
EXACT CODE → depends on dataset and threshold	RULE → cite source, period and convention	EXACT CODE → depends on dataset and threshold	RULE → cite source, period and convention

SYSTEM / DEFINITION

• COMMON TYPES → tropical, dry, subtropical, highland

• EXACT CODE → depends on dataset and threshold

PROCESS / MECHANISM

• BOUNDARY → depends on scale and interpolation

• RULE → cite source, period and convention

SPATIAL / INDIA PATTERN

• COMMON TYPES → tropical, dry, subtropical, highland

• EXACT CODE → depends on dataset and threshold

HAZARD / LIMIT / RESPONSE

• BOUNDARY → depends on scale and interpolation

• RULE → cite source, period and convention

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXACT CODE → depends on dataset and threshold.

09

STAGE 09CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISONTEMPERATURE + PRECIPITATION THRESHOLDSMODIFIED THERMAL ZONATIONWATER BALANCE + PETMATCH METHOD TO QUESTION PURPOSE

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose

CLASSIFICATION

• KOPPEN → temperature + precipitation thresholds

• TREWARTHA → modified thermal zonation

DIAGNOSTIC BASIS

• THORNTHWAITE → water balance + PET

• CHOICE → match method to question purpose

PROCESS LINK

• KOPPEN → temperature + precipitation thresholds

• TREWARTHA → modified thermal zonation

EXAM TRAP

• THORNTHWAITE → water balance + PET

• CHOICE → match method to question purpose

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09

STAGE 09

CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISON

TEMPERATURE + PRECIPITATION THRESHOLDS

MODIFIED THERMAL ZONATION

WATER BALANCE + PET

MATCH METHOD TO QUESTION PURPOSE

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose

CLASSIFICATION

• KOPPEN → temperature + precipitation thresholds

• TREWARTHA → modified thermal zonation

DIAGNOSTIC BASIS

• THORNTHWAITE → water balance + PET

• CHOICE → match method to question purpose

PROCESS LINK

• KOPPEN → temperature + precipitation thresholds

• TREWARTHA → modified thermal zonation

EXAM TRAP

• THORNTHWAITE → water balance + PET

• CHOICE → match method to question purpose

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THORNTHWAITE → water balance + PET; CHOICE → match method to question purpose.

10

STAGE 10

PLANNING OVERLAY

PLANNING OVERLAY

CLIMATE TYPE

LONG-PERIOD THERMAL-RAIN PATTERN

CROP AND SEASON CONTEXT

SOIL + LANDFORM + GROWING PERIOD

RISK LAYER

HAZARDS, EXPOSURE AND VULNERABILITY

CLIMATE TYPE → long-period thermal-rain pattern

AGRO-CLIMATE → crop and season context

AGRO-ECOLOGY → soil + landform + growing period

RISK LAYER → hazards, exposure and vulnerability

SYSTEM / DEFINITION

• CLIMATE TYPE → long-period thermal-rain pattern

• AGRO-CLIMATE → crop and season context

PROCESS / MECHANISM

• AGRO-ECOLOGY → soil + landform + growing period

• RISK LAYER → hazards, exposure and vulnerability

SPATIAL / INDIA PATTERN

• CLIMATE TYPE → long-period thermal-rain pattern

• AGRO-CLIMATE → crop and season context

HAZARD / LIMIT / RESPONSE

• AGRO-ECOLOGY → soil + landform + growing period

• RISK LAYER → hazards, exposure and vulnerability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AGRO-ECOLOGY → soil + landform + growing period.

11

STAGE 11

CLIMATE ANSWER SPINE

CLIMATE ANSWER SPINE

PURPOSE, VARIABLES AND PERIOD

LETTERS AND THRESHOLDS

INDIA WITH TRANSITION CAVEAT

METHOD, USE, LIMITATION AND VERDICT

DEFINE → purpose, variables and period

DECODE → letters and thresholds

MAP → India with transition caveat

COMPARE → method, use, limitation and verdict

DEFINITION / COMPONENTS

• DEFINE → purpose, variables and period

• DECODE → letters and thresholds

TRIGGER / MECHANISM

• MAP → India with transition caveat

• COMPARE → method, use, limitation and verdict

EFFECT / INTERVENTION

• DEFINE → purpose, variables and period

• DECODE → letters and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → India with transition caveat; COMPARE → method, use, limitation and verdict.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

A CLIMATES IN INDIA INCLUDE TROPICAL HUMID CLIMATES; AM

B CLIMATES ARE DRY

BSHW SEMI-ARID STEPPE AND BWHW HOT DESERT.

CWG IS THE GANGES/NORTH INDIAN PLAINS MONSOON CLIMATE WITH

H/E APPLIES TO THE HIMALAYA ABOVE HIGH ELEVATIONS/SNOW-LINE CONDITIONS.

TREWARTHA MODIFIES KOPPEN AND INCLUDES HIGHLAND CLIMATE FOR INDIA.

THORNTHWAITE USES PRECIPITATION–EVAPOTRANSPIRATION MOISTURE BALANCE, NOT TEMPERATURE ALONE.

OPTIONAL PROCESS DEPTH

• A climates in India include tropical humid climates; Am relates to heavy monsoon rain, while Aw marks savanna/dry-winter interior.

• B climates are dry: BShw semi-arid steppe and BWhw hot desert.

• Cwg is the Ganges/north Indian plains monsoon climate with dry winter.

• H/E applies to the Himalaya above high elevations/snow-line conditions.

DATA / INSTITUTION LIMIT

• Trewartha modifies Koppen and includes highland climate for India.

• Thornthwaite uses precipitation–evapotranspiration moisture balance, not temperature alone.

• India has Koppen A, B, C and H in standard applied schemes.

• Am/Amw = West Coast and very wet monsoon belt; Aw = peninsular savanna interior.

CONTEMPORARY PARALLEL

• As = Tamil Nadu/Coromandel dry-summer tropical type linked to NE monsoon rainfall.

• BWhw = hot desert of western Rajasthan/Thar/Kachchh.

• BShw = semi-arid steppe/rain-shadow and NW margins.

• Cwg = Ganges type: north Indian plains, monsoon with dry winter.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.